Lecture 14

# The toric code: topology protects

**Objectives.** *Lecture 13 ended with a question:* can protection grow with the size of the system? *The toric code answers it. Its qubits sit on the edges of a square lattice on a torus, each star and plaquette acts on four qubits, and it encodes two qubits (Theorem 14.2). Its distance is the size $L$ (Theorem 14.3). Its errors are strings, and their syndromes, at the ends, are anyons (Theorem 14.4). This is the second half of the third answer, and the door to Part II.*

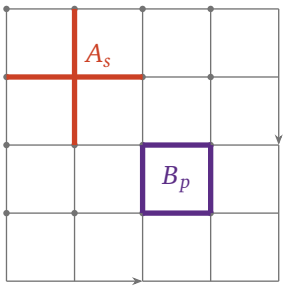
By the end of this lecture you will be able to:

- Define the toric code, and prove that its stars and plaquettes commute and that it encodes $k = 2$.
- Show that contractible loops and dual loops are products of stabilizers and that the straight loops around the torus are logical, and state that $d = L$.
- Prove that errors are strings whose syndromes sit at their ends, that taking an $e$ around an $m$ gives $-1$, and that the logical $X$ and $Z$ anticommute for the same reason.
- Say what "protected" means: what least-weight decoding corrects on $L = 3$, and how failure falls as $L$ grows, under a stated model.

## 14.1 The lattice, commutation

Take $L \geq 2$ and the $L \times L$ square lattice on a torus: vertices $(i, j)$ with $i, j \in \mathbb{Z}_L$ (row $i$, column $j$), the horizontal edge $h(i, j)$ from $(i, j)$ to $(i, j+1)$, and the vertical edge $v(i, j)$ from $(i, j)$ to $(i+1, j)$, with all indices modulo $L$: the right side is glued to the left, the bottom to the top. There are $L^2$ vertices, $L^2$ faces and $2L^2$ edges. The face $p(i, j)$ has the edges $h(i, j)$, $h(i+1, j)$, $v(i, j)$ and $v(i, j+1)$. As $L \geq 2$, the four edges at a vertex are distinct, so are the four of a face, and each edge has two distinct ends and borders two distinct faces.

**Definition 14.1** (The toric code)**.** Put a qubit on each of the $2L^2$ edges. The *star* of a vertex $s$ is $A_s = \prod_{e \ni s} X_e$, over the four edges at $s$; the *plaquette* of a face $p$ is $B_p = \prod_{e \in \partial p} Z_e$, over the four edges of its boundary. The *toric code* is the code $C(S)$ of the group $S$ generated by all stars and plaquettes.



The star of $s = (1, 1)$ and the plaquette of $p = p(2, 2)$ on $L = 4$; the arrows mark the sides that are glued. Stars commute with stars, and plaquettes with plaquettes. A star and a plaquette share no edge unless $s$ is a corner of $p$, and then exactly the two edges of $\partial p$ at $s$: they differ on 0 or 2 qubits, so they commute (§12.1). So $S$ is abelian; that $-\mathbb{1} \notin S$ is proved below. This is Kitaev's definition (1997, eq. (1), p. 2), and Dennis, Kitaev, Landahl and Preskill's (p. 4); Kitaev writes $k$, and Kitaev, Shen and Vyalyi write $r$ (p. 170), for our $L$.

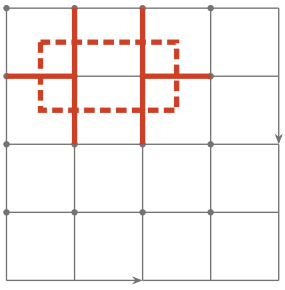
A set $c$ of edges is a *loop* if every vertex meets an even number of its edges, and a *dual loop* if every face has an even number of them on its boundary: the *dual edges*, one across each edge of $c$ joining the two faces at it, then form closed paths through the faces. As the stars are products of $X$'s and the plaquettes of $Z$'s, a product $X(a)Z(b) = \prod_{e\in a} X_e \prod_{e\in b} Z_e$, times a phase, commutes with every star exactly when $b$ is a loop, and with every plaquette exactly when $a$ is a dual loop.

## 14.2 $k = 2$

**Theorem 14.2** ($k = 2$)**.** *For $L \geq 2$, a product of stars and plaquettes is $\pm\mathbb{1}$ only if it uses all the stars or none and all the plaquettes or none, and it is then $\mathbb{1}$; so $-\mathbb{1} \notin S$. Dropping one star and one plaquette leaves $2L^2 - 2$ independent generators of $S$, and the toric code has dimension 4: $k = 2$.*

*Proof.* Let $T$ be a set of vertices. Each edge lies in the stars of its two ends, and $X_e^2 = \mathbb{1}$, so $\prod_{s\in T} A_s = X(a)$, with $a$ the set of edges that have exactly one end in $T$. If $a$ is empty, every neighbour of a vertex of $T$ is in $T$, and as the lattice is connected, $T$ is empty or all of it. Likewise, for a set $Q$ of faces, $\prod_{p\in Q} B_p = Z(b)$, with $b$ the set of edges that border exactly one face of $Q$, and $b$ is empty only if $Q$ is empty or all of it, as the faces are connected across edges. Stars commute with plaquettes, so $\prod_T A_s \prod_Q B_p = X(a)Z(b)$, in any order and with no phase. It is a multiple of $\mathbb{1}$ only if $a$ and $b$ are empty: a qubit in $a$ or $b$ carries $X$, $Z$ or $XZ = -iY$. So the only products that are $\pm\mathbb{1}$ are the four with $T$ and $Q$ each empty or all, and each is $\mathbb{1}$.

Now drop $A_{s_0}$ and $B_{p_0}$. If a remaining generator were a product of the others, a product of stars and plaquettes with $s_0 \notin T$, $p_0 \notin Q$ and $T$ or $Q$ nonempty would be $\mathbb{1}$, which we excluded. And the dropped two are products of the rest, $A_{s_0} = \prod_{s\neq s_0} A_s$ and $B_{p_0} = \prod_{p\neq p_0} B_p$, so the rest still generate $S$. By Theorem 12.3 the code has dimension $2^{2L^2-(2L^2-2)} = 4$. ∎



The stars of $(1, 1)$ and $(1, 2)$ share $h(1, 1)$, which cancels: their product is $X$ on the six other edges at the two vertices, the edges that the dashed dual loop around them crosses. A contractible dual loop is a product of stars.

The two relations $\prod_s A_s = \prod_p B_p = \mathbb{1}$ make the $2L^2$ generators dependent, as were Lecture 12's $Z_1Z_2$, $Z_2Z_3$, $Z_1Z_3$: counting them all would give $k = 0$. Kitaev (p. 3), Dennis et al. (p. 5: "There are no further relations among these operators"), Kitaev, Shen and Vyalyi

(p. 171) and Watrous (2025, p. 450) state that these are the only relations; none of them proves it. The proof uses only that the lattice and its faces are connected.

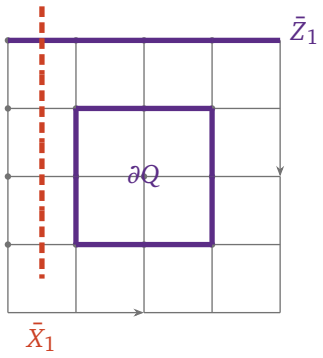
## 14.3 Loops as logical operators; $d = L$

**Theorem 14.3** (Logical operators · *(iii) cited*)**.** *(i) $Z$ on the boundary of a set of faces is the product of their plaquettes, and $X$ on the edges leaving a set of vertices is the product of their stars: such loops act as $\mathbb{1}$ on the code.*

*(ii) The loops $z_1 = \{h(0,j)\}_j$ and $z_2 = \{v(i,0)\}_i$ around the torus, and the dual loops $x_1 = \{h(i,0)\}_i$ and $x_2 = \{v(0,j)\}_j$, give $\bar{Z}_a = Z(z_a)$ and $\bar{X}_a = X(x_a)$ in $N(S) \setminus \langle i\mathbb{1}\rangle S$, of weight $L$, with $\bar{Z}_a\bar{X}_b = (-1)^{\delta_{ab}}\bar{X}_b\bar{Z}_a$.*

*(iii) The distance is $L$.*

*Proof of (i) and (ii).* (i) is in the proof of Theorem 14.2. (ii) Each vertex meets $z_a$ in 0 or 2 edges, and each face meets $x_a$ in 0 or 2, so the four commute with $S$. $z_1$ and $x_1$ share the edge $h(0,0)$, $z_2$ and $x_2$ the edge $v(0,0)$, and $z_1$, $x_2$ and $z_2$, $x_1$ share none. An element $cg$ of $\langle i\mathbb{1}\rangle S$ commutes with every element of $N(S)$; $\bar{Z}_a$ anticommutes with $\bar{X}_a \in N(S)$, so it is not in $\langle i\mathbb{1}\rangle S$, nor is $\bar{X}_a$. ∎




A contractible loop, the boundary of the four faces $Q$ it encloses, is the product of their plaquettes. $z_1$ and $x_1$ run around the torus and cross once. A loop that crosses $x_1$ an odd number of times, or $x_2$, cannot be the boundary of faces: it goes around the torus (loops modulo boundaries: $\mathbb{Z}_2$ homology, named here, not built).

Part (iii) is cited: every element of $N(S) \setminus \langle i\mathbb{1}\rangle S$ has a part, its $X$-string or its $Z$-string, that goes around the torus, and meets each of $L$ disjoint rows or columns (Kitaev, p. 4; Dennis et al., pp. 5–6; Kitaev, Shen and Vyalyi, p. 171; Watrous, 2025, p. 456). It is checked for $L = 2$, 3 and 4. The sources pair the logical operators differently (Kitaev, p. 8; Kitaev, Shen and Vyalyi, p. 175); here a pair is a loop and a dual loop that cross once.

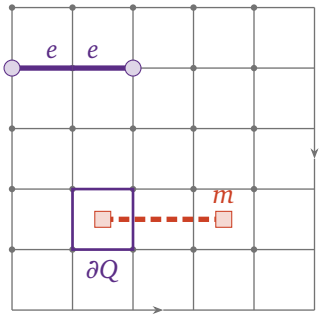
## 14.4 Strings, anyons, $-1$

An error $X(a)$ violates the plaquettes $p$ that meet $a$ in an odd number of edges, and $Z(b)$ the stars that meet $b$ in an odd number; a $Y$ is both. Call a violated star an $e$ and a violated plaquette an $m$. A *path* is a sequence of edges, each starting where the last ended; a *dual path* is a sequence of dual edges, from face to face.

**Theorem 14.4** (Strings and anyons). *(i) $Z$ on a path from a vertex $u$ to a vertex $w \neq u$ violates exactly the stars at $u$ and $w$; $X$ on a dual path from a face $f$ to a face $g \neq f$ violates exactly the plaquettes $f$ and $g$.*

*(ii) Let $\psi$ be in the code, $t$ a dual path from $f$ to $g \neq f$, and $Q$ a set of faces with $f \in Q$ and $g \notin Q$. Then $Z(\partial Q)\, X(t)\psi = -X(t)\psi$.*

*Proof.* (i) Each passage of the path through a vertex uses two edges at it, and the two ends one each; an edge used twice cancels, as $Z_e^2 = \mathbb{1}$. So the star of a vertex meets the path in an odd number of edges exactly at $u$ and $w$. The dual case is the same, faces for vertices. (ii) By Theorem 14.3(i), $Z(\partial Q) = \prod_{p \in Q} B_p$, and by (i), $B_p$ anticommutes with $X(t)$ exactly for $p = f$ and $p = g$; one of them is in $Q$. So $Z(\partial Q)X(t) = -X(t)Z(\partial Q)$, and $Z(\partial Q)\psi = \psi$. ∎




So errors are strings, and their syndromes sit at the ends: $Z$ on a path leaves two $e$'s (above), $X$ on a dual path two $m$'s (below). Read (ii) as a process: an $e$ pair is made by $Z$ on one edge of $\partial Q$, one $e$ is taken once along $\partial Q$, around the region $Q$ that holds the $m$ at $f$, and the pair is removed; the net operator is $Z(\partial Q)$, and the state changes sign. In the picture $\partial Q$ crosses the $m$-string once. The logical $\bar{X}_a$ and $\bar{Z}_a$ anticommute for the same reason: their loops cross once. This is Kitaev's computation (§2, Fig. 4, p. 7; $W = -1$, p. 8); also Kitaev, Shen and Vyalyi (p. 175) and Dennis et al. (p. 9). Kitaev's "$z$-type particles" are our $e$'s.

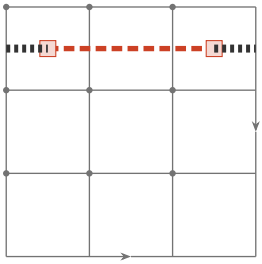
The $-1$ is a monodromy: an $e$ taken once around an $m$. It is not the phase of an exchange, as $e$ and $m$ are different particles; de Resende's reading of it as "spin-$\frac{1}{4}$" (p. 17) says more than it does. A phase other than +1 for one particle taken around another is what makes $e$ and $m$ anyons, which two space dimensions allow (*Symmetries*, L15, a starred reading).

## 14.5 $L = 3$ and suppression

Take $L = 3$: 18 qubits and $d = 3$. Look at $X$ errors; $Z$ errors are the same on the dual lattice. An error $X(a)$ shows the $m$'s at the ends of its dual strings. *Least-weight decoding* applies an $X(b)$ of least weight with the same syndrome. Then $X(a)X(b) = X(a+b)$, with $a+b$ the edges in one of $a$, $b$ but not both, has no syndrome: $a+b$ is a dual loop. The correction succeeds if $X(a+b)$ is a product of stars, which acts as $\mathbb{1}$ on the code, and otherwise leaves a logical error.

Every single $X$ error is corrected: on $L = 3$ two faces share at most one edge, so the syndrome of one edge, the two faces at it, is that of no other single edge, and the decoder

undoes the error itself. Two errors can defeat it:



$X$ on $v(0,1)$ and $v(0,2)$ (dashed) shows $m$'s at $p(0,0)$ and $p(0,2)$; $p(0,1)$, which holds both, is not violated. The least-weight correction is the one edge between $p(0,2)$ and $p(0,0)$, across the glued sides: $X$ on $v(0,0)$ (dotted). The three make $X(x_2) = \bar{X}_2$, a logical error: the decoder closed the loop the short way round, and the error had gone the long way. With $d = 3$, one error is guaranteed to be corrected, not two. Of the 153 pairs of $X$ errors, 18 fail this way, the three pairs on each of the six straight dual loops; 81 have one least-weight correction, which succeeds, and 54 have several, which differ by stars and all succeed: errors that differ by a star act alike on the code, which is degenerate for them (§11.6).

"Protected" does not mean that failures are absent; it means that they are suppressed as $L$ grows, under a stated model of noise and decoding. Dennis et al. take independent $X$ and $Z$ errors on each qubit, each with probability $p$ per round. With perfect syndromes, a round of errors is corrected with a failure probability that tends to 0 as $L$ grows when $p$ is below about 10.5% with least-weight decoding, and below about 10.9% with optimal decoding (values of Kawashima and Aoki, and of Honecker et al., which they quote, pp. 17–18). When each syndrome bit is also wrong with probability $p$, independently across checks and rounds and of the data errors, they decode the history of $T$ rounds of syndromes at once, by least weight and with exact classical processing, and prove that for $p < 1.14\%$, a lower bound on the threshold, the failure probability decreases exponentially with $L$, for $T$ growing no faster than a polynomial in $L$ (pp. 10–11, 18–20).

## 14.6 The third answer

Protection can grow with the size of the system. Each star and each plaquette of the toric code acts on four qubits, whatever $L$, while every logical operator goes around the torus: $d = L$. An error does harm only when its strings, with the decoder's, close a loop around the torus, and below threshold the probability of that falls exponentially with $L$, under the models above (Kitaev, p. 5; Dennis et al., p. 20). This completes the third answer: errors are discrete and correctable (Lectures 10–11), stabilizer codes give them a working form (Lecture 12), a protected computation needs Clifford gadgets, magic states and a threshold (Lecture 13), and topology makes protection grow with size. The anyons here are abelian: taking $e$ around $m$ multiplies the state by a number.

**Summary.**

- *The toric code:* qubits on the $2L^2$ edges, commuting stars of $X$'s and plaquettes of $Z$'s; $k = 2$.
- *Loops:* contractible loops and dual loops are products of stabilizers; the straight loops around the torus are logical; $d = L$ (cited).
- *Anyons:* errors are strings with $e$'s or $m$'s at their ends; an $e$ taken around an $m$ gives $-1$, and $\bar{X}_a$, $\bar{Z}_a$ anticommute, as their strings cross once.

- *Protection:* on $L = 3$ one error is corrected, two can make a logical error; below threshold, failure falls exponentially with $L$, under a stated model.
- *Next:* Part II: anyons whose braiding acts by matrices, not numbers, compute by braiding.

## References

A. Yu. Kitaev, arXiv:quant-ph/9707021 (1997), pp. 2–8. E. Dennis, A. Kitaev, A. Landahl and J. Preskill, arXiv:quant-ph/0110143 (2001), pp. 4–6, 9–11 and 17–20. Kitaev, Shen and Vyalyi, pp. 170–175. Watrous (2025), pp. 446–458. M. F. de Resende, arXiv:1712.01258, p. 17.